

Failure Modes for ASG + ALB setup

1. Broken user data

- Symptom:
TG identified unhealthy instance
- Root cause:
Faulty user data on launch template used by ASG did not start / enable httpd → TG health check was setup to use HTTP
- Fastest detection signal:
CloudWatch metric notification via SNS or directly check ALB / TG
- Prevention:
Ensure launch template is standardized properly; enable ELB health checks on ASG; register instances as targets for TG

2. Max capacity too low

→ Symptom:

CPU usage gets throttled up; no new instances booted up

→ Root cause:

Desired capacity settings on ASG were suboptimal; in the real world this could also be bad Automatic scaling policies, use best configuration setup based on your use case.

→ Fastest detection signal:

New instances not booting up to support; again CloudWatch metric SNS fire off if ASG has all resources above 40% CPU util or other defined metric

→ Prevention:

Set realistic desired capacity based on use case; ensure reliable scaling policy

3. Wrong subnet or SG

→ Symptom:

Failures showing everywhere from where you blocked traffic. For me I deleted all inbound rules on the Instance - SG so the ALB and TGs could not perform health checks and the ALB didn't work anymore

→ Root cause:

Deleted inbound rules on SG

→ Fastest detection signal:

Flood of failures on ALB / TGs

→ Prevention:

Choose proper rules set; apply least privilege and only allow authorized people to touch it through IAM along policies